

MANAS PATHAK

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Professional Summary

Computer Science Engineer with hands-on experience in **Python, Java, SQL, machine learning, and AI/LLM systems**. Built production-grade platforms integrating **REST APIs, relational databases, and deep learning models**. Strong foundation in **data structures and algorithms, object-oriented programming, DBMS**, and **scalable backend architecture**. Experienced with **cloud deployment, version control, and blockchain smart contracts**.

Technical Skills

Languages: Python, Java, C/C++, Solidity, SQL

Data & ML: NumPy, Pandas, Scikit-learn, PyTorch, Feature Engineering, Data Cleaning, Transformers

AI / NLP / LLM: LangChain, LangGraph, RAG, Prompt Engineering, Embeddings, Multi-Agent Systems

Backend & Databases: REST APIs, FastAPI, PostgreSQL, MySQL, SQLAlchemy (Async), Query Optimization

Tools & Cloud: Git/GitHub, Linux, Docker, AWS, PyPI, Smart Contracts

Project Experience

TimeForge — *GNSS Clock Bias Prediction & Analysis System*

- Built a full-stack system to monitor, analyze, and predict **GNSS satellite clock bias and orbit errors** using time-series forecasting models.
- Developed a **Transformer-based deep learning model** (PyTorch) with **satellite embeddings**, achieving high-precision prediction of clock corrections.
- Implemented a React + Three.js **3D visualization dashboard** with an **AI agent** (LangChain + Gemini API) for natural-language satellite analytics.

Mapicx — *Lightweight Deep Learning Framework*

300+ downloads on PyPI

- Engineered a neural network framework from scratch in **Python and NumPy** with Dense, Dropout layers and **ReLU/Softmax** activations.
- Developed an automatic differentiation system with gradient clipping and normalization, improving **training convergence by 15%**.
- Designed a **Keras-like API** (compile, fit, predict) and published the package on **PyPI**, accumulating 300+ downloads.
- Achieved **96.7% accuracy** on a spiral dataset benchmark, validating model correctness.

MapR1 — *Autonomous Multi-Agent Dream Simulator Engine*

- Architected a **multi-agent dream simulation engine** in Python where LLM-powered agents autonomously strategize, form alliances, and generate emergent scenarios from a single natural-language prompt.
- Built a **Contextual Consequence Engine** that propagates causal impacts across agents, dynamically shifting trust scores, alliance stability, and world tension using non-linear weight scaling.
- Implemented async database layer (**SQLAlchemy + Supabase PostgreSQL**) with a **Scenario Seeder** pipeline that auto-generates factions, economies, hidden knowledge, and goal conflicts from seed prompts.
- Designed a **Dream Director** and **Story Compiler** that transforms raw simulation logs into chapter-by-chapter dream sequences with per-agent diary exports.

Education

JSPM's Rajarshi Shahu College of Engineering

Bachelor of Technology (B.Tech), Computer Science — 9.0 CGPA

Class 12 — **82.67%**

Class 10 — **92.60%**

Work Experience

Google Developer Group

Jul 2025 – Jun 2026

Technical Lead | Blockchain Team

- **System Design & Optimization:** Architected an ERC-721-compliant digital marketplace with wallet integration and cost optimization algorithms, reducing per-transaction overhead by 30%.
- **Team Leadership:** Led a cross-functional team to ship two end-to-end platforms (token system + marketplace), coordinating design, development, and campus-wide rollout.

Achievements

- **Finalist — Smart India Hackathon (SIH):** Built a **policy intelligence platform** using **RAG, pgvector, FastAPI, and role-based access control**, recognized nationally for scalable document intelligence and compliance automation.
- **Open Source (PyPI):** Published **Mapicx**, a deep learning framework on **PyPI** with **300+ downloads** and a companion technical article, achieving **96.7% benchmark accuracy** via optimized SGD training.
- **DSA / Problem Solving:** Solved **250+ coding problems** across **LeetCode, GeeksforGeeks, and NeetCode**, strengthening fundamentals in data structures, algorithms, and competitive programming.